Ryan Swift

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RESEARCH FOCUS

- **Representation Learning**: Knowledge distillation, mechanistic interpretability, world models
- **Science of LLMs**: interpretability, multimodal reasoning, agentic alignment
- **Applied Interpretability**: performance, efficiency, alignment, safety

EDUCATION

University of Southern California May 2029 (Expected)
PhD in Computer Science *Advisor: Konstantinos Psounis*

- Researching multimodal representation learning and its applications
- Interests include world models, interpretability, safety, and alignment
- Relevant Coursework: Trustworthy Foundation Models, High Dimensional Mathematics, Advanced NLP

University of California, Davis June 2024
B.S. in Computer Science *Advisors: Zubair Shafiq, Chen-Nee Chuah*

- Undergraduate researcher in the University Honors Program
- Designed a novel ML authentication pipeline, achieving 90% accuracy on user data
- Relevant Coursework: Optimization, Computer Security, Software Engineering, Theory of Computation

WORK EXPERIENCE

Graduate Student Intern at LLNL CSI Summer 2026
Mentor: Colin Ponce

- Focused on integrating agentic workflows into critical infrastructure such as power systems as part of the *Stormbreaker* project
- Contributed to building custom agentic harness, as well as conducted independent research for agentic safety and alignment
- Doubled feature set of harness, including tools for agents to interact with threat intelligence databases
- Researched how open-source models respond when attacked with prompt injections designed to induce sycophancy (paper coming soon)

RESEARCH EXPERIENCE

RAPID: Randomized Projections for Internal Distillation August 2026–Present
Collaborators: Konstantinos Psounis

- Distillation of internal activations for improving reasoning in ¡1B parameter models
- Transfers geometric representation of pairwise relationships
- Applies JL Lemma for data-independent projection that preserves structure
- Reduces training costs by as much as 35.2% without compromising performance

SAE Feature-based Optimizations for Instruction Tuning January 2026–Present
Collaborators: Konstantinos Psounis

- Interpretable method for pruning/filtering/compressing datasets
- Captures task-relevant human-interpretable features
- Leverages architecture of sparse autoencoders (SAEs) to derive feature importance
- Demonstrates utility in instruction-tuning setting
- Achieves comparable performance to full training using only 100 out of 50K+ samples

KDFP: Knowledge Distillation from First Principles August 2025–Present
Collaborators: Konstantinos Psounis

- First principles exploration of white-box knowledge distillation
- Evaluation of over 100 novel distillation methodologies
- Demonstrates as much as 2.5% (0-shot) and 1.5% (5-shot) performance improvement over comparable methods
- Reduces training data requirement from 32B tokens to 100M without additional parameters

CPU-Auth: Hardware Authentication via Side-Channel Analysis March 2023–June 2024
Collaborators: Hari Venugopalan, Kaustav Goswami, Kartik Patwari, Chen-Nee Chuah, Zubair Shafiq

- Trained ResNet, MLP, and CNN pipelines which achieved up to 90% accuracy on real-world data
- Uncovered hardware-based fingerprint via time-series analysis of CPU side-channel
- Compiled report and code documentation to hand off to research group upon graduation
- Directed entire project group in final semester

TECHNICAL PROFICIENCIES

ML Frameworks: Pytorch, TensorFlow, HuggingFace Transformers, TransformerLens, SAE Lens
Languages: Python (NumPy, Pandas, Sci-Kit Learn, SciPy), C/C++, Java, Bash, Lua, Nix
Tools: Git, Nix, Ray, Pytorch-Lightning, Weights & Biases, Docker
Methods: LoRA, Finetuning, Knowledge Distillation, Representational Similarity Analysis

SELECTED PROJECTS

Fully Reproducible System and Home Configurations with Nix December 2025–Present
github.com/ryanaswift7/nix-dots

- Binary-reproducible configuration of NixOS system
- Portable Home Manager configuration works even on non-NixOS systems
- Hierarchical, dendritic organization

LoRA Fine-tuning Implementation September 2024–December 2024
github.com/ryanaswift7/gpt2_LoRA_567

- Reproduced original LoRA paper results, then implemented 6.85% improvement over baseline